

Blake McAlister

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EDUCATION

Clemson University

BS in Computer Science, Minor in Artificial Intelligence (GPA: 3.3/4.0)

Clemson, SC

Graduation Date: December 2026

SKILLS

Programming: Python, Java, JavaScript, C/C++, SQL, HTML

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Computer Vision (YOLO, OpenCV, OCR), NLP, VLMs

Data Science & Analytics: NumPy, Pandas, Data Cleaning, Exploratory Data Analysis, Statistical Analysis, Data Visualization

Full-Stack Development: AWS, Flask, FastAPI, React, REST APIs, API Integration, Docker, Node.js, Backend Development

Systems: Git/GitHub, Linux/Unix, Unit & Integration Testing, High-Performance Computing (GPU, Palmetto Cluster)

EXPERIENCE

Capgemini

Clemson, SC

Industry-Sponsored AI Capstone

August 2026 - Present

- Developing ShopBot Commerce, a cloud-native, multi-agent retail platform on Azure that connects browsing, purchase, review, and support data to enable autonomous customer recommendations and engagement.
- Leading development of the customer data layer, designing ingestion pipelines and unified customer data models that provide real-time context to recommendation, promotion, sentiment, and engagement agents.
- Implementing Azure OpenAI, Azure AI Search, and event-driven Azure services to support agent orchestration, retrieval-augmented generation (RAG), workflow automation, analytics, and gated human intervention.

Trimble Inc.

Atlanta, GA

Software Engineering Intern

May 2026 - August 2026

- Designed and built an Electron.js desktop prototype integrating multiple Trimble web applications into a unified native shell with isolated application sessions and dynamic app registration.
- Debugged and shipped production fixes across the Trimble Pay monorepo, driving changes through GitHub pull requests, code reviews, testing, and established release workflows.
- Engineered Gainsight PX analytics across Trimble Pay workflows, implementing frontend selectors and resolving multi-route tracking conflicts through wildcard URL matching and AND/OR exclusion logic.

Watt AI Creative Inquiry Research

Clemson, SC

Audiogram Deep Learning Researcher

August 2025 - December 2025

- Built a robust end-to-end ML pipeline integrating YOLO detection, coordinate calibration, and preprocessing to digitize handwritten clinical audiograms with high reliability.
- Engineered advanced OCR + Vision-LLM extraction systems to automatically interpret speech scores and output structured, Health Record ready data.
- Led deployment and documentation of framework, using Python, PyTorch/TensorFlow, OpenCV, Git, and HPC workflows.

Oldcastle APG

Atlanta, GA

Digital Solutions Intern

May 2025 - August 2025

- Integrated a chatbot into Oldcastle's B2B e-commerce platform to reduce manual support load by automating order tracking, FAQs, and customer support workflows.
- Assisted with backend improvements and API integrations within the BigCommerce ecosystem to support scalable commerce operations.
- Analyzed user behavior using Lucky Orange to identify friction points, improve site performance, and guide UX and UI.

RELEVANT PROJECTS

Urba

October 2025

- Built a real-time environmental alert platform at the NASA Space Apps Hackathon using NASA EONET and public APIs.
- Implemented a social-feed system for live mobile app, location-based reporting of fires, air quality, and extreme weather.
- Engineered a city-level Wellness Score covering U.S. cities by aggregating live environmental and user-reported data.

Vision-Based Autonomous RC Car

November 2025

- Implemented real-time lane detection + heading correction using computer vision on a TurboPi system.
- Added car-following logic: distance estimation, stopping behavior, and restart logic for safe trailing.
- Achieved stable tracking in real hallway tests and multi-lane simulation environments.

Stock Price Predictor

January 2025

- Built a Streamlit app using Python ML/data libraries, Yahoo Finance API retrieval, and moving-average visualizations.

EXTRACURRICULARS

ColorStack (Vice President). National Society of Black Engineers (NSBE). AI Club. Clemson WBB Practice Player. Club Basketball. Dae Han Black Belt.

Study Abroad: Mexico (May 2024): Merida, Campeche, Chichen Itza, Valladolid, Puerto Morelos.